

Enhancing Motor Imagery EEG Signal Classification with Simplified GoogLeNet

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Abstract—The brain-computer interface is a novel tool for interaction between the brain and external devices, and is an important tool at the forefront of international brain science research. Motor imagery does not require external stimulation and is one of the most widely used BCI paradigms, and is now widely used to address the treatment of neurological impairments such as paralysis, stroke, and acromegaly. Electrophysiological source imaging is a technique based on the observation of electromagnetic signals from the cerebral cortex, reflecting the electrophysiological activity of brain nerve cells on the scalp surface of the brain. The wavelet transform, on the other hand, is commonly used for the processing of non-smooth signals and aims to extract excellent signal features. This paper combines these two algorithms as feature extraction methods. In recent years, deep learning has been increasingly combined with MI-EEG-based BCI. GoogLeNet is one of the most classical convolutional neural networks and has made a splash in ImageNet image recognition. However, using GoogLeNet to train the BCI public dataset BCI IV Ila and dataset HGD has shown problems such as long training time, overfitting, and gradient disappearance. In response, this paper proposes an improved and streamlined version of GoogLeNet for training based on this dataset to achieve an overall improvement in accuracy, training time, number of parameters, and other indicators. It is useful for the realistic application of MI-based BCI systems.

Keywords—brain computer interface, motor imagery, EEG, GoogLeNet.

I. INTRODUCTION

A brain computer interface (BCI) is a direct connection between the human brain and an external device that does not rely on nerves or muscles [1]. BCIs can extract signals from the brain's neural activity and then decode them into control commands to control external devices such as robots or computers [2]. For example, subjects can control the movement of robotic arms and legs by imagining the movement of their hands and legs. This technology can, in principle, repair any problems with the brain or even reshape

its function, which is of great importance in the field of medical rehabilitation. It facilitates our understanding of the brain system and improves the functioning of the central nervous system. At the same time, this technology pushes the pace of human-machine integration and promises to eventually enable human-human and human-machine mind interaction and mind downloading, thus creating enormous economic, and social value [3].

BCI studies are based on three basic paradigms, namely motor imagery (MI), steady-state visual evoked potential (SSVEP), and event-related potentials (ERPs) [4]. Of these, the motor imagery paradigm does not require external stimulation and is based on the stimulation effect on specific areas of the brain when the subject imagines different physical activities but does not actually produce movement, and is the most popular paradigm for brain-computer interfaces [5]. Raw EEG signals are non-smooth and contaminated with artifacts from many sources of non-physiological and physiological factors, making feature extraction of great importance to the field of brain-computer interfaces. Therefore, this paper combines excellent feature extraction methods: source computation of ESI, and cwt wavelet transform aiming to improve the classification accuracy of the system.

For the classification of EEG signals, networks such as Alexnet, Vgg, and GoogLeNet can be used. GoogLeNet is the first basic neural network to use a parallel architecture instead of stacked convolutional layers, which can reduce computational and memory costs to a certain extent. However, the depth of GoogLeNet does not match the EEG data of the subjects. Using GoogLeNet to classify the publicly available dataset BCI IV Ila and dataset HGD presented problems of overfitting, long training times, and disappearing gradients, so we simplified and changed the GoogLeNet network by retaining the Inception base module while adding several base modules. We use the simplified GoogLeNet to train and test on two public datasets, and achieve certain improvements in accuracy, time, convergence speed, and the number of parameters.

II. METHODS

A. Datasets

In order to better validate the experimental effect of the simplified before-and-after network, two publicly available datasets are used as the basic datasets in this paper: the dataset BCI IV Ila from the BCI competition and the HGD public dataset. The public dataset BCI IV Ila has data for a total of nine subjects (called S1-S9). Each subject performed two

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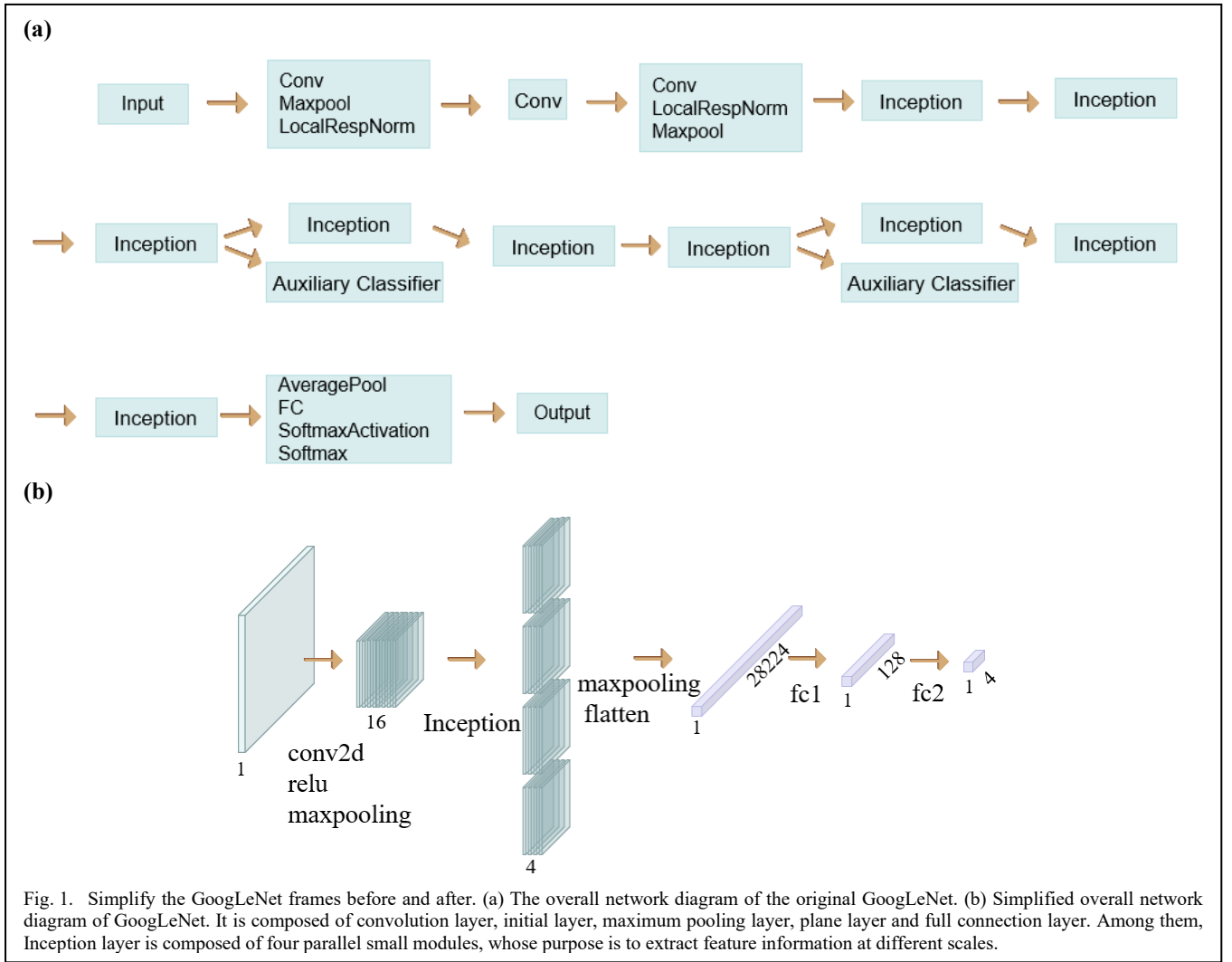


Fig. 1. Simplify the GoogLeNet frames before and after. (a) The overall network diagram of the original GoogLeNet. (b) Simplified overall network diagram of GoogLeNet. It is composed of convolution layer, initial layer, maximum pooling layer, plane layer and full connection layer. Among them, Inception layer is composed of four parallel small modules, whose purpose is to extract feature information at different scales.

motor imagery tasks on different days, a training phase and a testing phase. Each session consisted of 6 runs and each run consisted of 48 trails (including 12 left-handed, 12 right-handed, 12 bipedal, and 12 tongue motor imagery tasks), so the experiment contained a total of 576 motor imagery tasks. At the beginning of each session, a 5-minute baseline signal recording was recorded, i.e. gaze at the cross recording, eye closure recording, and eye movement recording. Each motor imagery session lasted 3 s. Scalp EEG was recorded at a sampling rate of 250 Hz on 10-20 electrodes in compliance with 22 international standards, with a total of 25 electrodes, including 23 EEG channels and 3 EOGs, with channel bandpass filtering of 0.5-100 HZ. For more detailed information about the public dataset BCI IV Ila, please visit https://www.bbci.de/competition/iv/desc_2a.pdf.

The HGD dataset recorded 1 resting task and 3 motor imagery tasks (left hand, right hand, and foot) for 14 healthy subjects (S01, S02, ..., S14), each lasting 4 s. The signals were recorded from 128 international standard 10-10 extended leads, sampled at 250 Hz. The HGD dataset further provided comparative data for the experiment Data. More detailed information about this dataset can be found at <https://gin.g-node.org/robintibor/high-gamma-dataset>.

B. Data Pre-processing

Raw EEG data are contaminated with artifacts from many non-physiological and physiological factors. These artifact components are identified and removed before the data are

analyzed, and processed, to prevent their impact on the experiment.

First, the scalp EEG is band-pass filtered at 6-32Hz to remove low-frequency drift and the interference of industrial frequencies to the experiment. The data were then subjected to a common average reference (CAR). The common average reference is kept constant concerning the EEG signal to reduce the effect of single point failures and to significantly reflect the difference between the EEG signal and the reference signal, facilitating the representation of brain activity relevant to the study. Given the realistic application scenario, no other pre-processing methods are applied to this dataset.

C. Electrophysiological Source Imaging Technology

Electroencephalography (EEG) is the classic method of recording brain activity using physiological indicators and is a general reflection of the electrophysiological activity of brain nerve cells on the scalp surface of the brain. However, the poor spatial resolution of EEG makes it difficult to infer the brain regions that generate neural activity based on the EEG signal [6]. Therefore, this paper converts the acquired EEG signals into a constant coefficient matrix of cortical EEG by combining head anatomical information and source localization algorithms. At the same time, the deficiency of the volume conduction effect was addressed by using electrophysiological source imaging (ESI), which effectively improved spatial resolution [7].

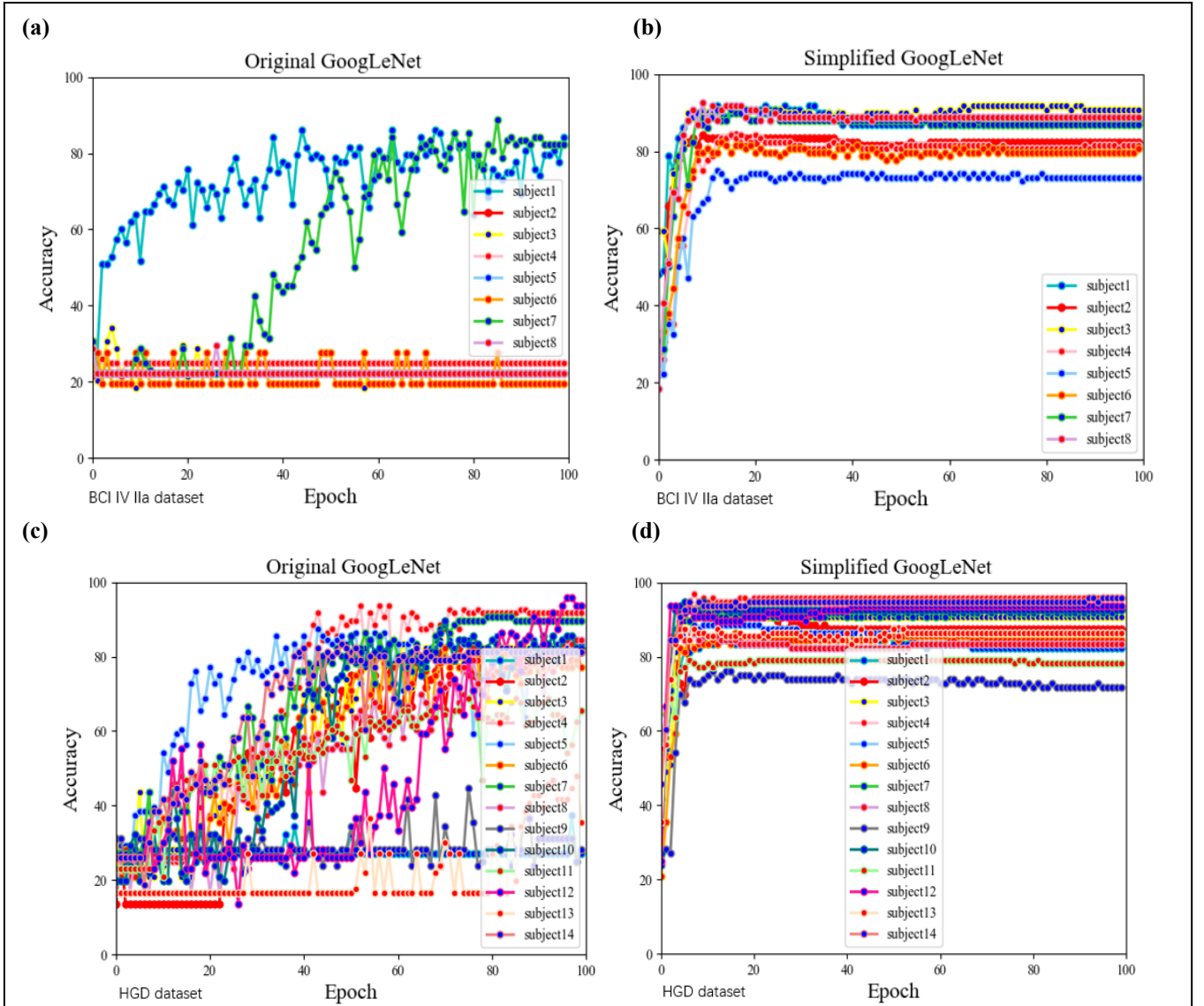


Fig.2. BCI IV Ila dataset and HGD dataset iterate accuracy graphs of 100 epochs on the original GoogLeNet and simplified GoogLeNet. (a) Training results of the BCI IV Ila dataset using the original GoogLeNet. (b) Training results of the BCI IV Ila dataset using the simplified GoogLeNet. (c) Training results of the HGD dataset using the original GoogLeNet. (d) Training results of the HGD dataset using the simplified GoogLeNet.

Firstly, ICBM152 (averaged results from 9 parameters of 152 MRI images of normal subjects after affine transformation matched with MNI.) The MRI images were processed to obtain a three-layer head model. The lead field matrix was then obtained according to the boundary element method, and the lead field matrix was able to effectively represent the process of EEG propagation changes within the brain. Since a large number of different source distributions can produce the same potential field on the scalp, this poses a "non-uniqueness" problem [8]. That is, the same scalp electrode measurement may produce different source spatial pattern signals. The conventional approach is to estimate the most likely source current distribution by some constraint method. This can be a mathematical problem as well as a biomedical and anatomical knowledge of the distribution of neuronal activity. The source solution provided depends on the degree of fit to the constraint. Finally, in contrast to minimum parametric estimation, this paper maps the EEG signal to a cortical EEG constant coefficient matrix using a dynamic parametric statistical mapping (dSPM) approach using both a noise covariance matrix and a normalization process [9].

D. Continuous Wavelet Transform (CWT)

Because motion imagines that EEG signals tend to change abruptly in the time or frequency domain to some degree, they are non-stationary signals. The Fourier transform cannot reflect the change in frequency over time [10]. The wavelet transform not only avoids the drawback that the window size does not vary according to frequency but also has the feature of multi-resolution analysis [11]. Therefore, this paper aims to generate a high-resolution spectral feature image by performing a time-frequency analysis with Morlet wavelet. The basic mathematical form of the Morlet wavelet is shown in equation (1), where w_0 denotes the central frequency [12].

$$\psi(t) = e^{(i\omega_0 t)} e^{-\frac{t^2}{2}} \quad (1)$$

The wavelet basis function is translated and telescoped to obtain equation (2): where a denotes the displacement value and b denotes the scale parameter.

$$\Psi f(a, b) = \int_{-\infty}^{+\infty} f(t) e^{-i\omega_0 \frac{(t-b)}{a}} e^{-\frac{(t-b)^2}{2a^2}} dt \quad (2)$$

Modify the values of the parameters and perform the analysis under different parameters. Obtain the time-

frequency information of the source signal $Q(s)$ (equation 3)

$$Q(s) = ||W_{s(b,a)}|| \quad (3)$$

TABLE I. THE BCI IV IIA DATASET LOOPS THROUGH THE ORIGINAL GOOGLNET AND SIMPLIFIED GOOGLNET FOR 5 ROUNDS (EACH ROUND INCLUDES 100 EPOCHS) WITH A COMPARISON TABLE OF AVERAGE ACCURACY AND STANDARD DEVIATION.

Subject	Original GoogLeNet	Simplided GoogLeNet
	Accuracy	Accuracy
S01	70.4%±25.9%	90.7%±3.4%
S02	22.2%±0.0%	82.6%±2.4%
S03	22.2%±0.0%	90.9%±0.8%
S04	25.0 %±0.0%	85.2%±2.5%
S05	22.2%±0.0%	66.9%±1.2%
S06	19.4%±0.0%	76.9%±2.5%
S07	33.3%±24.85%	85.0%±2.7%
S08	22.2%±0.0%	85.7%±1.8%
S09	72.4%±29.64%	90.2%±1.8%
mean	34.4%±8.9%	83.8%±2.1%

TABLE II. THE HGD DATASET LOOPS THROUGH THE ORIGINAL GOOGLNET AND SIMPLIFIED GOOGLNET FOR 5 ROUNDS (EACH ROUND INCLUDES 100 EPOCHS) WITH MEAN ACCURACY AND STANDARD DEVIATION COMPARISON TABLES.

Subject	Original GoogLeNet	Simplided GoogLeNet
	Accuracy	Accuracy
S01	64.2%±21.4%	84.0%±1.6%
S02	78.8%±12.3%	87.9%±1.6%
S03	70.8%±28.2%	91.9%±1.7%
S04	91.3%±2.8%	95.0%±2.0%
S05	84.0%±1.6%	88.3%±1.9%
S06	81.5%±2.4%	85.0%±2.4%
S07	84.0%±2.2%	92.1%±1.2%
S08	82.7%±2.4%	86.5%±1.9%
S09	28.3%±0.5%	76.7%±1.7%
S10	90.2%±2.6%	94.2%±1.6%
S11	64.6%±9.6%	78.3%±3.2%
S12	86.9%±3.7%	93.3%±1.4%
S13	38.8%±30.6%	88.1%±1.2%
S14	82.1%±0.9%	94.2%±2.7%
mean	73.4%±8.7%	88.2%±1.9%

E. Improved GoogLeNet Classification Model

1) GoogLeNet:

GoogLeNet is a network model architecture based on the Inception module proposed by Christian Szegedy in the ImageNet image recognition challenge in 2014 [13]. In general, the most straightforward way to improve the performance of a neural network model is to increase the

number of layers and width of the network, but blindly doing

TABLE III. THE BCI IV IIA DATASET ON THE ORIGINAL GOOGLNET AND SIMPLIFIED GOOGLNET ON THE AVERAGE TRAINING TIME OF A SINGLE PARTICIPANT AND GOOGLNET BEFORE AND AFTER THE SIMPLIFICATION OF THE PARAMETER NUMBER COMPARISON TABLE.

(BCI IV IIA)	Original GoogLeNet	Simplided GoogLeNet
Time(s)	773.12	151.88
Parameter quantity	5,971,380	3,616,612

TABLE IV. THE HGD DATASET COMPARES THE AVERAGE TRAINING TIME SPENT BY A SINGLE PARTICIPANT ON THE ORIGINAL GOOGLNET AND THE SIMPLIFIED GOOGLNET AND THE NUMBER OF PARAMETERS BEFORE AND AFTER GOOGLNET SIMPLIFICATION.

(HGD)	Original GoogLeNet	Simplided GoogLeNet
Time(s)	659.71	138.59
Parameter quantity	5,971,380	3,616,612

so can lead to many problems such as:

- The network is too deep and the complexity of the computation is too high.
- The amount of parameters in the network is too large leading to overfitting, making the accuracy of the training set much greater than that of the test set [14].
- The network is too deep, leading to the problem of gradient disappearance.

This paper proposes an improved and streamlined version of GoogLeNet, which retains the Inception method of the original GoogLeNet and achieves an overall improvement in the accuracy, training time, and the number of participants for training based on this dataset.

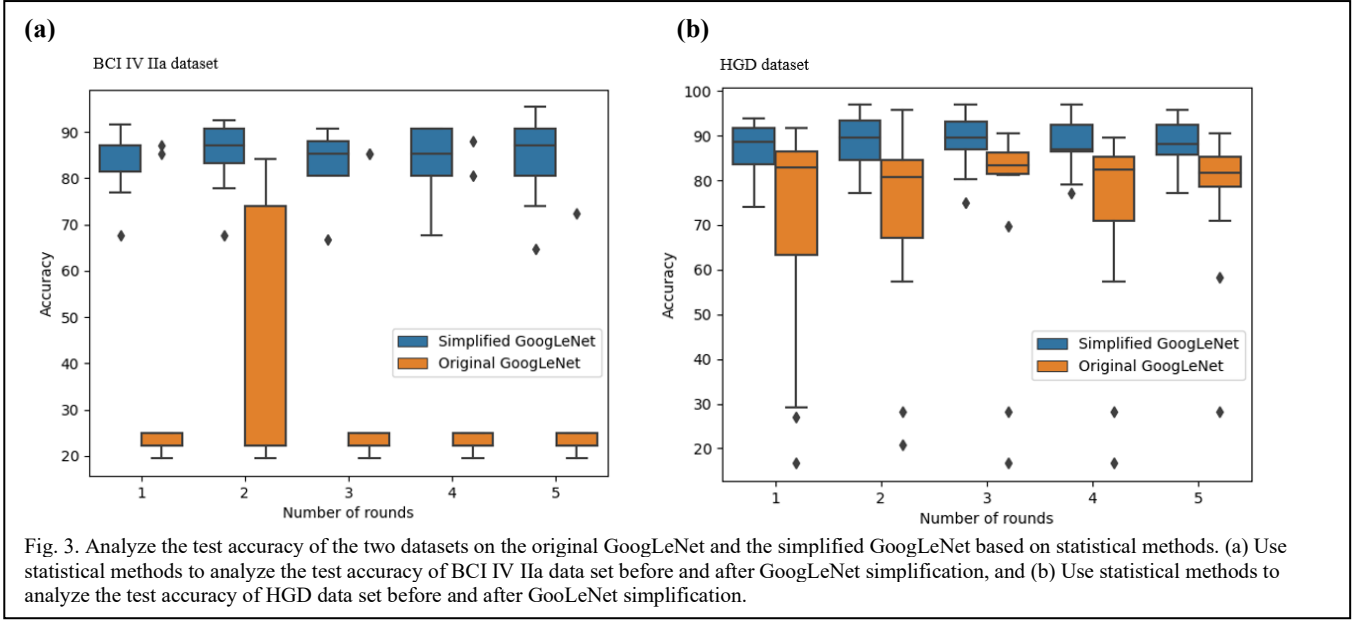
2) Inception layer:

In the network architecture without Inception, the size of the convolution kernel of the convolution layer is fixed and can only extract features of one size. However, in practical applications, using convolution kernels of different sizes has the best effect, because the features extracted by convolution kernels of different sizes are different, and so are the sizes of receptive fields. The Inception module can achieve this effect [15].

The Inception module consists of four small modules in parallel. The first three sub-modules extract feature information at different scales using convolutional kernels of sizes 1*1, 3*3, and 5*5 respectively. The purpose of using 1*1 convolutional kernels is to reduce the number of parameters and change the number of channels before inputting convolutional layers of kernel sizes 3 and 5. The fourth sub-module uses a 3×3 maximum pooling layer followed by a 1×1 convolutional layer to change the number of channels. Finally, the output of the four sub-modules is stitched together as the output of the Inception module [16].

3) Activation function and loss function:

An activation function is one in which the activated features are retained and mapped out as the input to the next neuron. If no activation function is used, then the output of each layer of neurons shows a linear relationship with the



input of the previous layer [17]. Clearly, no matter how many layers of neurons there are, the output of the entire neural network is a linear combination of the inputs, which is no different from having no hidden layers. The introduction of non-linear functions as activation functions gives more meaning and possibilities to neural networks so that they can approximate arbitrary functions. In this paper, the non-linear ReLU function is introduced as the activation function (Equation 4). Compared to other common activation functions, the ReLU function has the following advantages [18].

$$f(x) = \max(0, x) \quad (4)$$

a) The ReLU activation function is less computationally intensive than other activation functions.

b) The ReLU activation function can overcome the problem of gradient disappearance. [19].

c) ReLU does not activate neurons with negative inputs, which reduces parameter dependencies and drives the network to be sparse, which can mitigate the risk of overfitting [20].

The loss function uses the CrossEntropyLoss function, and the basic mathematical form is shown in Equation (5). Cross-entropy uses the concept of entropy, first converting the model into a value of entropy, and then using the value to compare the differences between models.

$$L = -[y \log \hat{y} + (1 - y) \log (1 - \hat{y})] \quad (5)$$

4) Overall Network Architecture:

The input data is output through the simplified network (Figure 1(b)). Compared with the network before simplification (Figure 1(a)), the simplified network has fewer layers and greatly reduced complexity.

Let the input tensor be $\mathbf{a}^i$, and the basic mathematical form of convolution is shown in Equation (6). Among them: superscript represents the number of layers; The subscripts s and i represent the number of convolution kernels in the convolutional layer and the number of times the convolution operation is performed, respectively; w and b represent the weights and biases of the convolution kernel, respectively.

$$\mathbf{a}_s^{(i+1)}(i) = f(\mathbf{a}_i^{(i)} \cdot \mathbf{w}_s^{(i+1)} + b_s^{(i+1)}) \quad (6)$$

In the training process, we used the Adam optimizer and the learning rate of 0.001. The computer was configured with the Inter Core i5-12400 CPU, an NVIDIA GeForce RTX 3050 GPU, and 64G of RAM.

III. RESULTS

In this part, we compare the performance of the original GoogLeNet and the simplified GoogLeNet from three dimensions of accuracy, time, and reference number. We found that compared with the performance of the original GoogLeNet, the simplified GoogLeNet has better performance in terms of accuracy, time, and reference quantity.

A. Comparison of Accuracy and Convergence Speed

We used GoogLeNet before and after simplification to train two kinds of public data sets respectively, and the accuracy line graph is shown in Figure 2. It can be seen that the accuracy of the data set using the original GoogLeNet training is low, and there are problems of gradient disappearing and slow convergence.

Five rounds of repeated training with 100 epochs per round were performed on the data from nine subjects in the BCI IV Ila dataset to obtain average accuracy (Table 1) and to plot a box plot according to statistical methods (Figure 3 (a)). Using the same method for 14 subjects in the HGD dataset, the average accuracy (Table 2) and boxplot (Figure 3 (b)) can be obtained. We can see: The average accuracy of GoogLeNet for BCI IV Ila data before and after simplification was 34.4% and 83.8%, respectively, and the average accuracy of GoogLeNet for HGD data before and after simplification was 73.4% and 88.2%, respectively. The accuracy and performance of the simplified network have been significantly improved.

B. Comparison of Training Time and Number of Parameters

In this section, we compare the number of parameters of the two networks and the time consumed to iterate 100 epochs per subject. As shown in table 3 and table 4, it can be noticed that there is a degree of improvement in both the time and the number of parameter dimensions.

IV. DISCUSSION

We hypothesize that the EEG data is a finite single-dimensional discrete time series of many sample points with a simple structure. Moreover, the EEG data already has a clear EEG signature after a series of excellent algorithms such as ESI, cwt, etc. have been used for feature extraction. Using a deeper neural network would not only cause an increase in computation time and a waste of memory resources, but it may also be the case that the shallow layers keep not learning and the later layers keep learning (i.e., the gradient disappears). The network fails to converge and the deeper layers are lost. Thus, for EEG signals, the original GoogLeNet depth is too deep and several problems arise. The Inception layer reduces the depth of the network by using different sizes of convolutional kernels to extract features while not using a strategy of stacking convolutional layers. The simplified GoogLeNet retains the Inception layer and adds a small number of base network layers, allowing features to be extracted from different dimensions while significantly reducing the depth of the network, achieving good performance on both public datasets.

V. CONCLUSION

The BCI IV Ila dataset achieved an average accuracy of 83.79% for 9 subjects on a simplified version of GoogLeNet, with a maximum accuracy of 90.9% for subjects, an average time spend of 151.88s for 100 epoch iterations for one subject, and a network participant count of 32,259,940. Meanwhile, the HGD dataset achieved an average accuracy of 87.28% across 14 subjects on a simplified version of GoogLeNet, with a maximum subject accuracy of 95.0%, an average time spend of 138.59s for a subject iterating through 100 epochs, and a network participant count of 32,259,940. These results show that the simplified version of GoogLeNet has good performance and contributes to the portability of the BCI system.

REFERENCES

- [1] R. A. M., "Adaptive transfer learning-based multiscale feature fused deep convolutional neural network for EEG MI multiclassification in brain-computer interface," *Engineering Applications of Artificial Intelligence*, vol. 116, 2022.
- [2] T. Fang *et al.*, "Decoding motor imagery tasks using ESI and hybrid feature CNN," (in English), *J Neural Eng*, vol. 19, no. 1, Feb 1 2022, doi: ARTN 016022 10.1088/1741-2552/ac4ed0.
- [3] "Ethics - Medical Ethics; Data on Medical Ethics Reported by Researchers at Ludwig-Maximilians-University Munich (What Is It Like To Use a BCI? - Insights From an Interview Study with Brain-computer Interface Users)," *Computer Technology Journal*, 2020.
- [4] H. Gou *et al.*, "A solution to supervised motor imagery task in the BCI Controlled Robot Contest in World Robot Contest," *Brain Science Advances*, vol. 8, no. 02, pp. 153-161, 2022.
- [5] I. Shun, N. Tsuyoshi, S. Shinya, A. Ryohei, K. Tomoyoshi, and O. Yukari, "Motor imagery enhances corticospinal transmission mediated by cervical premotoneurons in humans," *Journal of neurophysiology*, vol. 124, no. 1, 2020.
- [6] J. Komal, U. Rahul, and S. H. Shankar, "A novel EEG channel selection and classification methodology for multi - class motor imagery - based BCI system design," *International Journal of Imaging Systems and Technology*, vol. 32, no. 4, 2022.
- [7] S. Rui, S. Abbas, W. G. A., and H. Bin, "Deep neural networks constrained by neural mass models improve electrophysiological source imaging of spatiotemporal brain dynamics," *Proceedings of the National Academy of Sciences of the United States of America*, vol. 119, no. 31, 2022.
- [8] S. Martin, C. Lucia-Manuela, H. Mauritius, S. Thibaut, V.-V. Veerle, and M. C. M., "Subcortical electrophysiological activity is detectable with high-density EEG source imaging," *Nature communications*, vol. 10, no. 1, 2019.
- [9] B. He, A. Sohrabpour, E. Brown, and Z. Liu, "Electrophysiological Source Imaging: A Noninvasive Window to Brain Dynamics," *Annual Review of Biomedical Engineering*, vol. 20, no. 1, 2018.
- [10] A. Enas *et al.*, "Resting state EEG-based diagnosis of Autism via elliptic area of continuous wavelet transform complex plot," *Journal of Intelligent & Fuzzy Systems*, vol. 39, no. 6, 2020.
- [11] X. Tang, W. Li, X. Li, W. Ma, and X. Dang, "Motor imagery EEG recognition based on conditional optimization empirical mode decomposition and multi-scale convolutional neural network," *Expert Systems With Applications*, vol. 149, no. C, 2020.
- [12] N. Mammone, C. Ieracitano, and F. C. Morabito, "A deep CNN approach to decode motor preparation of upper limbs from time-frequency maps of EEG signals at source level," *Neural Networks*, vol. 124, no. C, 2020.
- [13] Yang *et al.*, "A Convolutional Neural Network of GoogLeNet Applied in Mineral Prospectivity Prediction Based on Multi-source Geoinformation," *Natural Resources Research*, vol. 30, no. 6, 2021.
- [14] W. Satoru and Y. Hayato, "Overfitting measurement of convolutional neural networks using trained network weights," *International Journal of Data Science and Analytics*, vol. 14, no. 3, 2022.
- [15] W. Xuan *et al.*, "SpikeGoogle: Spiking Neural Networks with GoogLeNet - like inception module," *CAAI Transactions on Intelligence Technology*, vol. 7, no. 3, 2022.
- [16] Pathak, Debanjan, Raju, and U. S. N., "Content-based image retrieval using Group Normalized-Inception-Darknet-53," *International Journal of Multimedia Information Retrieval*, vol. 10, no. 3, 2021.
- [17] D. S. Ram, S. S. Kumar, and C. B. Baran, "Activation functions in deep learning: A comprehensive survey and benchmark," *Neurocomputing*, vol. 503, 2022.
- [18] H. Juncai, L. Lin, and X. Jinchao, "ReLU deep neural networks from the hierarchical basis perspective," *Computers and Mathematics with Applications*, vol. 120, 2022.
- [19] H. Juncai, L. Lin, and X. Jinchao, "Approximation properties of deep ReLU CNNs," *Research in the Mathematical Sciences*, vol. 9, no. 3, 2022.
- [20] L. Min, C. Zhiqiang, and C. Jingshuang, "Adaptive two-layer ReLU neural network: I. Best least-squares approximation," *Computers and Mathematics with Applications*, vol. 113, 2022.